Linear Time Invariant (LTI) Systems

Semester 2, 2025/2026



Properties of LTI Systems

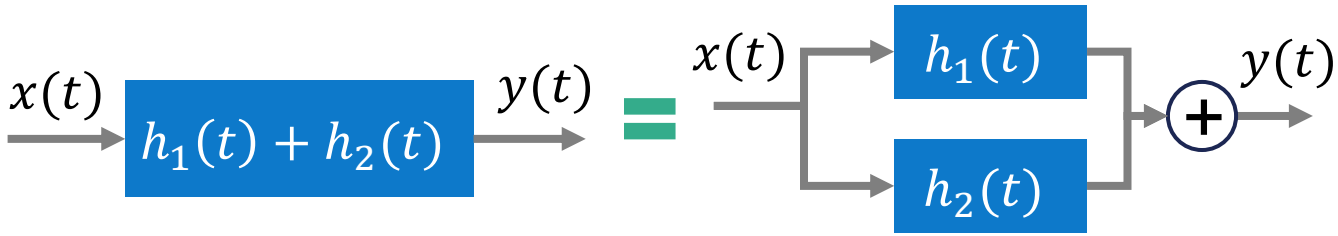
1. **Commutative Property**

   $x * h = h * x$



2. **Distributive Property**

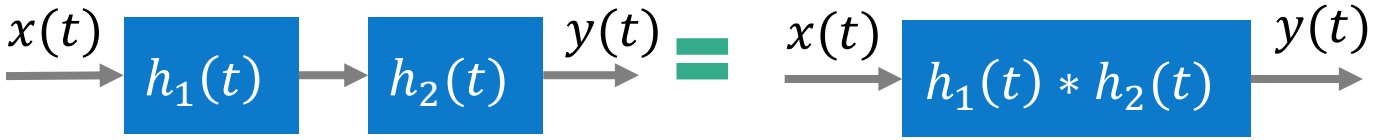
   $x * (h_1 + h_2) = x * h_1 + x * h_2$



LTI systems in parallel can be collapsed into a single LTI system

3. **Associative Property**

   $x * (h_1 * h_2) = (x * h_1) * h_2$



Example 2.6: Distributive Property

Let $y[n]$ denote the convolution of the following two sequences:

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + 2^n u[-n]$$

$$h[n] = u[n]$$

Find $y[n]$ using distributive property.

Solution:

Direct evaluation of such a convolution is somewhat tedious.

Let $x_1[n] = \left(\frac{1}{2}\right)^n u[n]$ and $x_2[n] = 2^n u[-n]$,

$$y[n] = y_1[n] + y_2[n] = x_1[n] * h[n] + x_2[n] * h[n]$$

Example 2.6: Distributive Property (cont'd)

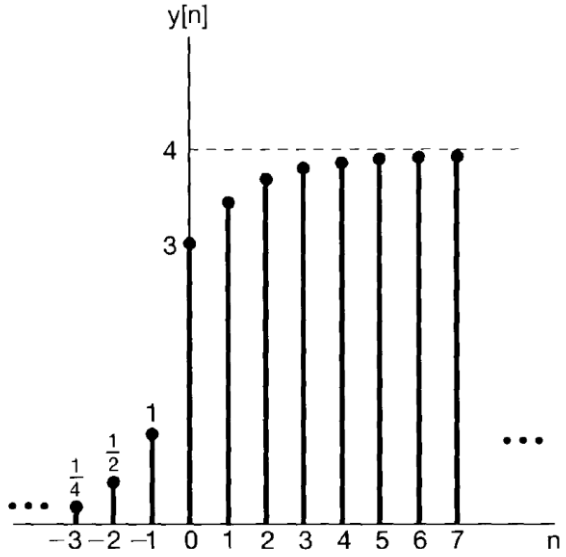
The convolution for $y_1[n]$ can be obtained from Example 2.2 with $\alpha = \frac{1}{2}$ while $y_2[n]$ was evaluated in Example 2.3.

$$y_1[n] = \left(\frac{1-(1/2)^{n+1}}{1-(1/2)}\right) u[n] = y_1[n] = (2 - (1/2)^n)\, u[n] \text{for all } n$$

$$y_2[n] = \begin{cases} 2, & n \geq 0 \\ 2^{n+1}, & n < 0 \end{cases}$$

Therefore,

$$y[n] = \begin{cases} 4 - (1/2)^n, & n \geq 0 \\ 2^{n+1}, & n < 0 \end{cases}$$



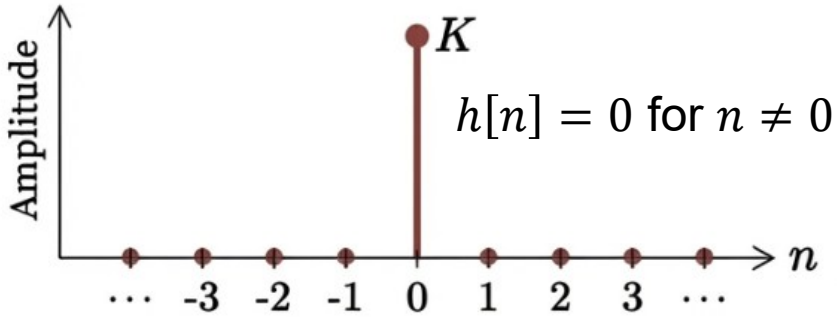
Properties of LTI Systems

4. **Memoryless Systems**

   Output depends only on current input.

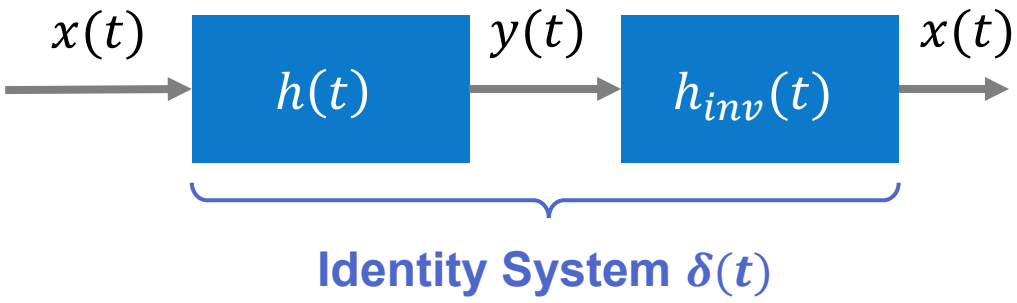
   $h[n] = K\delta[n]$

   $y[n] = Kx[n]$



5. **Invertible Systems**

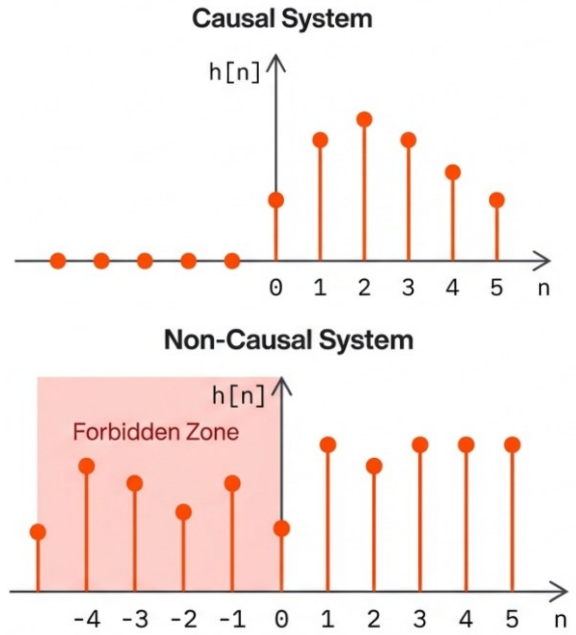
   Input can be recovered from output. (Condition: $h * h_{inv} = \delta$)



Properties of LTI Systems

6. Causality

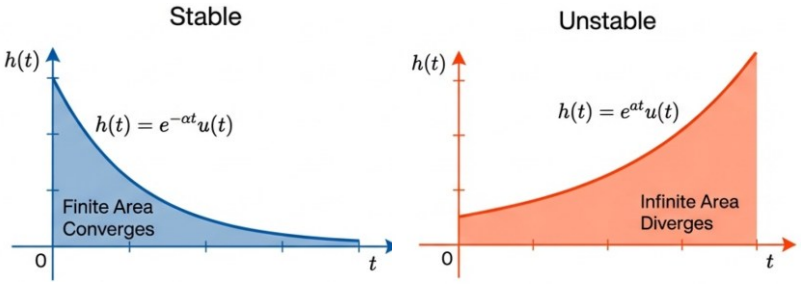
- The output of a causal system depends only on the present and past values of the input to the system.
- The impulse response of a causal LTI system must be zero before the impulse occurs, i.e. $h[n] = 0$ for all $n < 0$ (DT) or $h(t) = 0$ for all $t < 0$ (CT)



7. Stability

- A system is stable if every bounded input produces a bounded output (BIBO).
- Condition: The impulse response must be *absolutely summable* (DT) or *integrable* (CT).

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty \qquad \text{or} \qquad \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$



Causality for LTI Systems

Examples

- **Ideal predictor**: $y(t) = x(t+1)$ - noncausal since the output at time t depends on the input at future time $t+1$
- **Ideal delay**: $y(t) = x(t-1)$ - causal since the output at time t depends only on the input at past time $t-1$
- **Moving average (MA) filter**: $y[n] = \frac{x[n-1]+x[n]+x[n+1]}{3}$ - noncausal, since the output at time n depends in part on the input at future time $n+1$

Most physical systems are causal. However, noncausal systems are widely used in signal processing, for example, for smoothing of continuous-time and discrete-time signals for noise removal or quality enhancement.

Exercise

The following are the impulse responses of discrete-time/continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers.

a) $h[n] = \left(\frac{1}{5}\right)^n u[n]$

b) $h[n] = (0.8)^n\, u[n+2]$

c) $h[n] = \left(\frac{1}{2}\right)^n u[-n]$

d) $h(t) = e^{-4t}u(t-2)$

e) $h(t) = e^{2t}u(-1-t)$

Answer:

a) Causal, stable
b) Not causal, stable
c) Not causal, unstable
d) Causal, stable
e) Not causal, stable

Unit Step Response of an LTI System

The unit step response, $s[n]$ of a system is defined as

$$s[n] = u[n] * h[n]$$

(i.e., the step response of a system is the output it produces for a unit-step function input $x[n] = u[n]$).

$$s[n] = \sum_{k=-\infty}^{\infty} h[k]$$

$h[n]$ can be recovered from $s[n]$ using the relation

$$h[n] = s[n] - s[n-1]$$

Unit Step Response of an LTI System

Similarly, for continuous-time systems,

$$s(t) = u(t) * h(t)$$

The step response s and impulse response h of a LTI system are related as

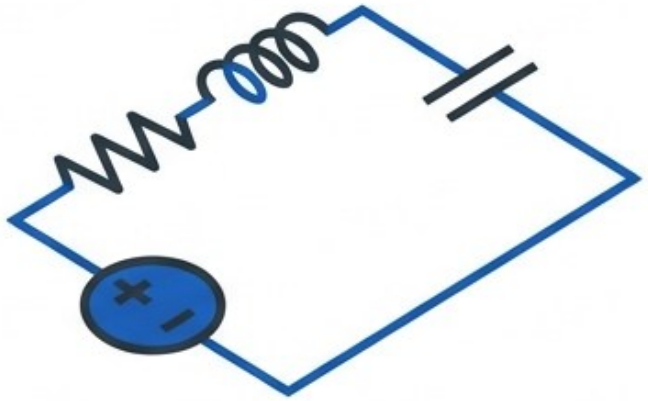
$$s(t) = \int_{-\infty}^{t} h(\tau) d\tau$$

and

$$h(t) = \frac{ds(t)}{dt}$$

Causal LTI Systems in Real World

Continuous Time



Differential Equations

Examples: RLC Circuits, Mechanical Springs
Nature: Smooth change over time

Discrete Time



Difference Equations

Examples: Digital Filters, Recursive Code
Nature: Step-wise sequences

Linear Constant-Coefficient Differential Equations

Modeling Continuous-Time LTI Systems

A general Nth-order linear constant-coefficient differential equation is given by

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

- The differential equation does not completely specify the output in terms of the input, and we need to identify **auxiliary conditions** to determine completely the input-output relationship for the system.
- The solution $y(t)$ consists of two parts: $\boldsymbol{y(t) = y_p(t) + y_h(t)}$
  - **Particular solution $y_p(t)$** - use the condition of initial rest (Assuming $x(t) = 0$ for $t < t_0$ implies $y(t) = 0$ for $t < t_0$ ensures causality and linearity)
  - **Homogeneous solution $y_h(t)$** (natural response)

Example: Continuous Time Analysis

Process	Math
System & Input	$\frac{dy(t)}{dt} + 2y(t) = x(t) \qquad x(t) = Ke^{3t}u(t)$
1. Find Particular Solution (y_p)	Substitute $y_p = Ye^{3t} \rightarrow 3Y + 2Y = K \rightarrow y_p(t) = \frac{K}{5}e^{3t}$
2. Find Homogeneous (y_h)	Root $s = -2 \rightarrow y_h(t) = Ae^{-2t}$
3. Apply Initial Rest	$y(0) = 0 \rightarrow 0 = A + \frac{K}{5} \rightarrow A = -\frac{K}{5}$
Final Solution	$y(t) = \frac{K}{5}\left[e^{3t} - e^{-2t}\right]u(t)$

Linear Constant-Coefficient Difference Equations

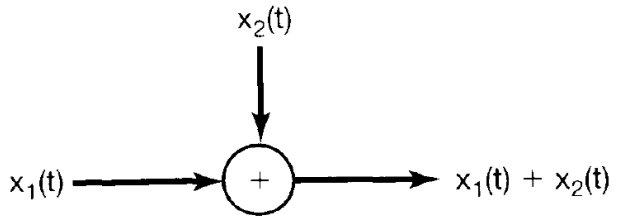
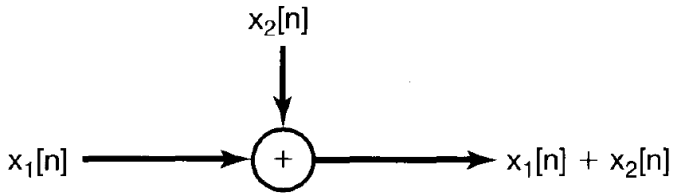
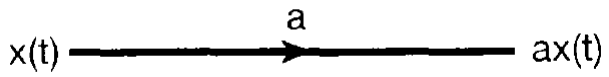
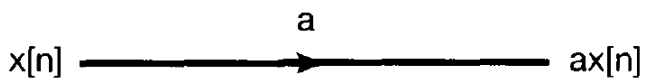
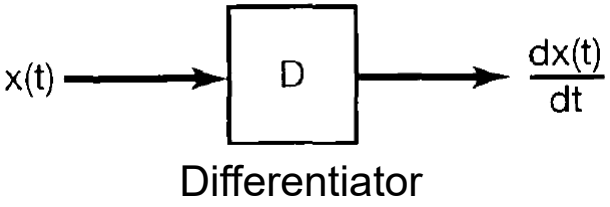
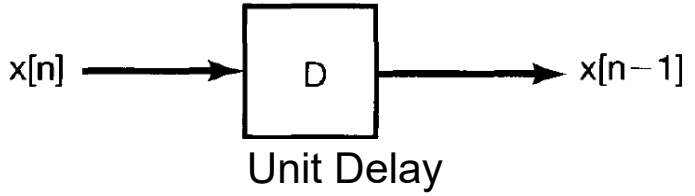
Modeling Discrete-Time LTI Systems

A general Nth-order linear constant-coefficient difference equation is given by

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

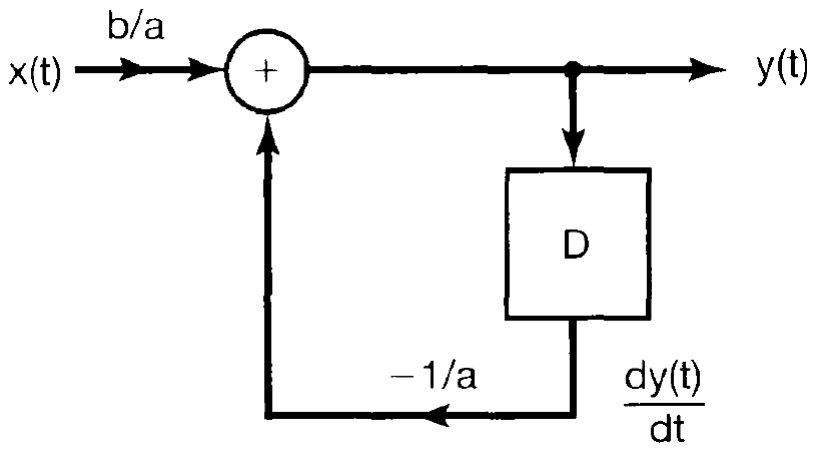
- An equation of this type can be solved in a manner exactly analogous to that for differential equations.
- The solution $y[n]$ can be written as the sum of a **particular solution** and a solution to the **homogeneous equation**.

Block Diagram Representations of First-Order Systems

Function	Visual Symbol (CT)	Visual Symbol (DT)
Adder		
Multiplier (by a coefficient)		
Unit Delay / Differentiator		

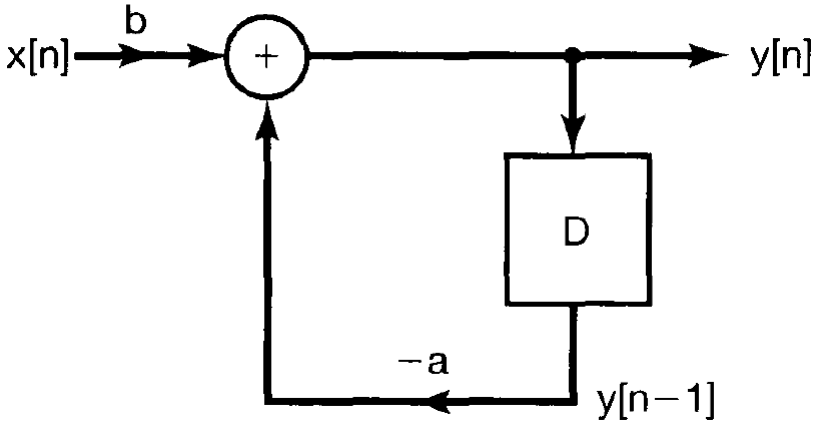
Example 2.7

Continuous-Time



$$\frac{dy(t)}{dt} + ay(t) = bx(t)$$

Discrete-Time



$$y[n] + ay[n-1] = bx[n]$$